

A Compact Narrative for Prediction-Relevant Operator Mechanisms

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1 The Question

The starting point is a simple ambiguity. A transformer trained for in-context linear regression may make predictions close to linear-kernel ridge regression, but pointwise accuracy on one sampled label vector does not identify the computation being used. Many algorithms can agree with ridge regression on the realized labels while responding differently to nearby counterfactual labels.

The note therefore changes the object of study. Instead of asking only whether the transformer predicts the right targets for one sampled context, it freezes the input geometry and perturbs only the context labels. The first question becomes:

When labels are changed infinitesimally, does the transformer respond like the ridge operator?

This operator-response test is only the entry point. The paper then asks a stronger mechanistic question: if the transformer has a ridge-like local response, when does a hidden layer contain a compact context-side subspace sufficient to reconstruct the ridge operator at the prediction-risk scale of the task? Causal use is then a separate intervention question: after a sufficient subspace has been identified, do interventions preserving or removing that subspace preserve or damage the transformer’s response?

The resulting claim is stronger than behavioral imitation but narrower than global equality. In the successful regimes, the transformer locally realizes the task-distributed ridge label-response operator, and hidden response spans contain low-dimensional subspaces whose induced reduced KRR operators recover exact KRR up to a predeclared prediction-risk tolerance. The number of required directions is computed before looking at activations. The locality matters: the tests concern the sampled geometry, the chosen perturbation scale, and especially task-label directions rather than all possible label perturbations.

2 Fixed Geometry Makes Ridge Regression an Operator

Fix one in-context episode with context and target inputs

$$X_{\text{ctx}} \in \mathbb{R}^{n_{\text{ctx}} \times d_x}, \quad X_{\text{tgt}} \in \mathbb{R}^{n_{\text{tgt}} \times d_x}.$$

The fixed geometry is

$$G = (X_{\text{ctx}}, X_{\text{tgt}}).$$

During the operator tests, G is held fixed. The context inputs do not move, the target inputs do not move, and the only varying object is the context-label vector

$$y = (y_1, \dots, y_{n_{\text{ctx}}}) \in \mathbb{R}^{n_{\text{ctx}}}.$$

Define

$$K = X_{\text{ctx}} X_{\text{ctx}}^\top, \quad K_t = X_{\text{tgt}} X_{\text{ctx}}^\top, \quad A = K + \sigma^2 I.$$

For linear-kernel ridge regression, the dual coefficients and target predictions are

$$\alpha^\star = A^{-1}y, \quad y^\star = K_t \alpha^\star = K_t A^{-1}y.$$

Thus, once G is fixed, ridge regression is a linear map from context labels to target predictions:

$$T_G := K_t A^{-1}, \quad T_G : \mathbb{R}^{n_{\text{ctx}}} \rightarrow \mathbb{R}^{n_{\text{tgt}}}.$$

The transformer defines a map on the same spaces:

$$F_\theta^G : \mathbb{R}^{n_{\text{ctx}}} \rightarrow \mathbb{R}^{n_{\text{tgt}}}, \quad F_\theta^G(y) = \hat{y}_\theta.$$

When the geometry is clear, write T and F . The central comparison is not merely $F(y) \approx Ty$ for one y . It is whether the local response of F to label perturbations matches the operator T .

3 Context-Label Directions, Local Operators, and Additivity

A context-label direction is a vector

$$u \in \mathbb{R}^{n_{\text{ctx}}}.$$

It assigns one scalar to each context example. Perturbing in direction u means changing labels as

$$y + \varepsilon u = (y_1 + \varepsilon u_1, \dots, y_{n_{\text{ctx}}} + \varepsilon u_{n_{\text{ctx}}}).$$

This is a direction over context positions. It is not a direction in input-feature space and not a direction in target space. If there are 47 context examples, these directions live in $\mathbb{R}^{47}$.

Pointwise agreement checks only the value of two maps at one label vector:

$$F(y) \approx Ty.$$

Since fixed-geometry ridge regression is linear, its Jacobian is the operator itself:

$$J_T(y) = T \quad \text{for every } y.$$

The transformer map F can be nonlinear in the labels. The weakest local version of “the same label-to-prediction rule” is therefore equality of first-order responses:

$$J_F(y)u \approx Tu.$$

Equivalently, for small label perturbations,

$$F(y + \delta) \approx F(y) + T\delta.$$

The experiments estimate the Jacobian action with the central finite difference

$$D_\varepsilon F(y)[u] := \frac{F(y + \varepsilon u) - F(y - \varepsilon u)}{2\varepsilon}.$$

The symmetric form is used because it cancels the leading first-order bias of a one-sided difference: for a smooth map, the error is $O(\varepsilon^2)$ rather than $O(\varepsilon)$. It also treats positive and negative label

perturbations in the same direction symmetrically, which is important when the goal is to estimate the local tangent response rather than a one-sided effect. The behavioral operator test asks whether

$$D_\varepsilon F(y)[u] \approx Tu$$

on held-out label directions.

This is still a local statement. Matching the value and first derivative at one point does not imply global equality. For example, $f(x) = x + x^2$ and $g(x) = x$ have the same value and derivative at $x = 0$, but differ away from zero. Thus derivative matching supports tangent-level agreement, not $F(y') = Ty'$ for all possible labels.

The probe distribution specifies which counterfactual label directions are tested. The main probes are task-label probes:

$$u \sim \mathcal{N}(0, A).$$

This is the natural law because, under the matched linear task

$$y = X_{\text{ctx}}\beta + \eta, \quad \beta \sim \mathcal{N}(0, I), \quad \eta \sim \mathcal{N}(0, \sigma^2 I),$$

conditioning on the fixed geometry leaves only β and η random, so

$$y \mid G \sim \mathcal{N}(0, X_{\text{ctx}}X_{\text{ctx}}^\top + \sigma^2 I) = \mathcal{N}(0, A).$$

Equivalently, the conditional label covariance is

$$\text{Cov}(y \mid G) = A.$$

Thus $u \sim \mathcal{N}(0, A)$ samples counterfactual label perturbations with the same covariance geometry as labels generated by the task, after the input geometry has been fixed. In this precise sense, task probes are the directions the task distribution itself emphasizes. In whitened coordinates, they are $u = A^{1/2}z$, $z \sim \mathcal{N}(0, I)$. The experiments also use isotropic probes

$$u \sim \mathcal{N}(0, I),$$

which test arbitrary context-label directions equally. The distinction is central: the claim is task-local because the transformer is close to ridge on task-label probes but not on arbitrary isotropic probes. Build probes are used to construct subspaces or fitted controls; held-out probes are used for reported errors.

For a held-out probe set $U = \{u_1, \dots, u_m\}$ and a fixed linear operator $S : \mathbb{R}^{n_{\text{ctx}}} \rightarrow \mathbb{R}^{n_{\text{tgt}}}$, define

$$E_\varepsilon(F, S; U) := \left(\frac{\sum_{j=1}^m \|D_\varepsilon F(y)[u_j] - Su_j\|_2^2}{\sum_{j=1}^m \|Tu_j\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

When $S = T$, this measures whether the transformer’s finite-difference response matches exact ridge regression. The denominator normalizes by the ridge-response energy on the same probes, so an error of 0.01 means a root-mean-square discrepancy of about one percent of the typical ridge response size.

Proposition 1 (Transformer errors are not bounded by 1). *For a finite probe set U , write*

$$R_U^2 = \sum_j \|Tu_j\|_2^2.$$

If $D_\varepsilon F(y)[u_j] = 0$ for all probes, then

$$E_\varepsilon(F, T; U) = \frac{R_U}{(R_U^2 + \eta_{\text{num}})^{1/2}} \leq 1.$$

If $D_\varepsilon F(y)[u_j] = -Tu_j$, then the same calculation gives a value at most 2. More generally, if $D_\varepsilon F(y)[u_j] = cTu_j$, then

$$E_\varepsilon(F, T; U) = |c - 1| \frac{R_U}{(R_U^2 + \eta_{\text{num}})^{1/2}},$$

so transformer-response errors have no universal upper bound.

Proof. Substitute the three proposed finite-difference responses into the definition of $E_\varepsilon(F, T; U)$. The last expression diverges as $|c| \rightarrow \infty$ whenever the ridge response energy R_U^2 is nonzero. $\square$

Because ridge is linear, its higher-order derivatives are zero. We do not try to estimate all higher-order derivatives of the transformer. Instead, we check whether nonlinear interactions are small at the tested perturbation scale. Ridge has

$$T(u + v) = Tu + Tv.$$

A nonlinear transformer need not satisfy

$$D_\varepsilon F(y)[u + v] \approx D_\varepsilon F(y)[u] + D_\varepsilon F(y)[v].$$

If this approximate additivity failed badly, then $E_\varepsilon(F, T)$ would mix two effects: failure to be ridge and failure to have a stable local linear response.

For paired directions $V = \{(u_j, v_j)\}_{j=1}^m$, define the second mixed finite difference

$$C_j = F(y + \varepsilon u_j + \varepsilon v_j) - F(y + \varepsilon u_j - \varepsilon v_j) - F(y - \varepsilon u_j + \varepsilon v_j) + F(y - \varepsilon u_j - \varepsilon v_j),$$

and the additivity defect

$$A_\varepsilon(F; V) := \left(\frac{\sum_{j=1}^m \|C_j\|_2^2}{\sum_{j=1}^m \|T(\varepsilon u_j + \varepsilon v_j)\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

For any affine function of the labels, $C_j = 0$ exactly. Because the defect is normalized by ridge-response energy, values below one mean the non-additive interaction is smaller than the typical ridge response on the tested paired probes; values around one mean it is ridge-scale; values above one mean the interaction is larger than the ridge response, so a local-linear interpretation is unreliable. This diagnostic does not identify the operator. It only says whether a local operator comparison is meaningful at the chosen ε and probe law.

4 Activation Subspaces and Prediction-Risk Rank

The previous section gives behavioral certificates:

$$F(y) \approx Ty, \quad D_\varepsilon F(y)[u] \approx Tu.$$

These are necessary, but they are still input-output facts. They do not say whether the transformer represents the directions needed for ridge regression, or whether those directions are used. The

mechanistic question is: does the residual stream expose a compact set of context-position directions sufficient for the prediction-relevant ridge computation?

This is the right internal object to look for because ridge uses context-position dual variables. For fixed geometry,

$$\alpha^*(y) = A^{-1}y \in \mathbb{R}^{n_{\text{ctx}}}, \quad Ty = K_t \alpha^*(y).$$

The dual coordinate α_i is indexed by the i -th context example. The context residual stream has the same context axis. At layer ℓ ,

$$H_\ell^{\text{ctx}} \in \mathbb{R}^{n_{\text{ctx}} \times D},$$

where D is the residual width. Rows are context tokens, and the k -th column $H_\ell^{\text{ctx}}[:, k]$ is a scalar pattern over context positions, hence a vector in $\mathbb{R}^{n_{\text{ctx}}}$. Linear combinations of residual-stream columns therefore define candidate directions in the same space as ridge dual variables.

Let

$$Q = [q_1, \dots, q_r] \in \mathbb{R}^{n_{\text{ctx}} \times r}$$

be a basis matrix for such candidate context-position directions. The subspace is the column span

$$\text{span}(Q) = \{Qc : c \in \mathbb{R}^r\} \subseteq \mathbb{R}^{n_{\text{ctx}}},$$

so Q is both a matrix representation and, by abuse of language, the subspace it spans. We use an A -orthonormal basis,

$$Q^\top A Q = I,$$

because the quadratic part of the ridge dual objective is $\frac{1}{2} \alpha^\top A \alpha$. Thus the natural size of a dual displacement is $\alpha^\top A \alpha$, and two directions are orthogonal in the ridge geometry when their cross-term in this quadratic energy vanishes.

Now define what it means for $\text{span}(Q)$ to be useful for ridge. This is a restriction of the dual solution, not an ordinary restriction of the input label vector. Full ridge solves

$$\alpha^*(y) = \arg \min_{\alpha \in \mathbb{R}^{n_{\text{ctx}}}} \left\{ \frac{1}{2} \alpha^\top A \alpha - \alpha^\top y \right\},$$

whose first-order condition is $A\alpha - y = 0$, hence $\alpha^* = A^{-1}y$. To restrict the dual solution to $\text{span}(Q)$, write $\alpha = Qc$. Plugging this into the same dual objective gives

$$\frac{1}{2} c^\top Q^\top A Q c - c^\top Q^\top y.$$

Since $Q^\top A Q = I$, this is

$$\frac{1}{2} c^\top c - c^\top Q^\top y.$$

Differentiating with respect to c gives $c - Q^\top y = 0$, so

$$c^* = Q^\top y.$$

Therefore the restricted dual solution is

$$\alpha_Q(y) = Q Q^\top y.$$

Without the A -orthonormal convention, the same calculation would give

$$\alpha_Q(y) = Q(Q^\top A Q)^{-1} Q^\top y.$$

Predictions are still read out through the context-target kernel matrix K_t , so the reduced ridge operator induced by Q is

$$T_Q := K_t Q Q^\top.$$

This is the bridge from residual-stream directions to an operator-level certificate. We are not fitting an arbitrary probe. We are asking whether the directions exposed by the residual stream are enough so that ridge, restricted to those directions, still predicts like full ridge.

The reason this sufficiency test is prediction-relevance weighted rather than global is that not every label direction matters equally for prediction. Conditional on the fixed geometry,

$$y \mid G \sim \mathcal{N}(0, A).$$

So the task naturally emphasizes perturbations

$$u \sim \mathcal{N}(0, A),$$

not arbitrary Euclidean directions. A direction is task-relevant when it is likely under this covariance geometry and when ridge produces a non-negligible target response along it. This leads to a purely ridge-theoretic benchmark: before looking inside the transformer, ask how many directions are needed by the best possible reduced ridge response on task-distributed labels.

To turn this into a prediction-scale benchmark, we need a reference risk. The target quantity is not the noisy context-label vector y . It is the vector of latent, noise-free function values at the target inputs. Write

$$f_{\text{tgt}} = (f(x_1^{\text{tgt}}), \dots, f(x_{n_{\text{tgt}}}^{\text{tgt}})) \in \mathbb{R}^{n_{\text{tgt}}},$$

and let K_{tt} be the target-target kernel matrix. In the linear case, $K_{tt} = X_{\text{tgt}} X_{\text{tgt}}^\top$. Under the matched GP/KRR model,

$$\begin{bmatrix} y \\ f_{\text{tgt}} \end{bmatrix} \mid G \sim \mathcal{N}\left(0, \begin{bmatrix} A & K_t^\top \\ K_t & K_{tt} \end{bmatrix}\right),$$

because $y = f_{\text{ctx}} + \eta$ includes label noise while f_{tgt} is the noise-free target function. Conditioning this Gaussian gives

$$\mathbb{E}[f_{\text{tgt}} \mid y, G] = K_t A^{-1} y = T y$$

and

$$\text{Cov}(f_{\text{tgt}} \mid y, G) = K_{tt} - K_t A^{-1} K_t^\top.$$

Thus exact KRR is the posterior mean predictor of the latent target values. It still has nonzero risk, because the noisy context labels do not determine the target function values exactly. Define this irreducible KRR posterior prediction risk by

$$R_{\text{KRR}}(G) := \mathbb{E} \|f_{\text{tgt}} - T y\|_2^2 = \text{tr}\left(K_{tt} - K_t A^{-1} K_t^\top\right),$$

where the expectation is conditional on the fixed geometry G .

Now compare exact KRR with any linear approximation S to T . Decompose

$$f_{\text{tgt}} - S y = (f_{\text{tgt}} - T y) + (T - S) y.$$

The first term is the posterior residual after exact KRR, and the second term is a function of the observed labels y . Since $T y = \mathbb{E}[f_{\text{tgt}} \mid y, G]$, the posterior residual is orthogonal to every function of y . Therefore the cross-term vanishes and

$$\mathbb{E} \|f_{\text{tgt}} - S y\|_2^2 = R_{\text{KRR}}(G) + \mathbb{E}_{y \sim \mathcal{N}(0, A)} \|(T - S) y\|_2^2.$$

The second term is the extra prediction risk caused by using S instead of exact KRR. Thus the natural scale for deciding whether a reduced operator is good enough is not zero operator error, but excess error relative to the irreducible risk already incurred by exact KRR. Define

$$\rho_G(S) := \frac{\mathbb{E}_{y \sim \mathcal{N}(0, A)} \|(T - S)y\|_2^2}{R_{\text{KRR}}(G) + \eta_{\text{num}}}.$$

This asks how much prediction risk is added by using S instead of exact KRR.

The next question is how many source directions are needed in principle. This should be answered before looking at the transformer. We therefore ask a purely KRR question: among all rank- r linear responses to task-distributed labels, how much extra prediction risk is unavoidable if we keep only r directions?

Whiten the task-label perturbations:

$$u = A^{1/2}z, \quad z \sim \mathcal{N}(0, I).$$

Here z is an isotropic coordinate system for labels drawn from the task covariance A . In these coordinates, the exact KRR response is

$$Tu = K_t A^{-1} A^{1/2} z = K_t A^{-1/2} z.$$

Define

$$C_G := K_t A^{-1/2}.$$

Thus C_G is not a new predictor. It is the same KRR operator written in coordinates where the task-label distribution is spherical. This distinction matters. Singular values of T measure response size to raw Euclidean label directions $u \sim \mathcal{N}(0, I)$, which is not the task law. Singular values of C_G measure response size to label directions that the task actually generates, $u \sim \mathcal{N}(0, A)$.

Let

$$C_G = U \text{diag}(s_i) V^\top, \quad s_1 \geq s_2 \geq \dots \geq 0.$$

If z points along the right singular vector v_i , the KRR prediction moves in the corresponding target direction with size s_i . Thus s_i^2 is the squared prediction-response energy carried by the i -th task-label mode. Since z is isotropic,

$$\mathbb{E} \|C_G z\|_2^2 = \sum_i s_i^2.$$

Now suppose we approximate exact KRR by a rank- r linear operator. In whitened coordinates this means approximating C_G by a rank- r matrix. By Eckart–Young, the best possible rank- r approximation keeps the r largest singular directions. The unavoidable task-law mean squared operator error is the discarded tail

$$\sum_{i>r} s_i^2.$$

Equivalently, the best possible rank- r predictor has total prediction risk

$$R_{\text{KRR}}(G) + \sum_{i>r} s_i^2.$$

This is why the tail is compared to $R_{\text{KRR}}(G)$, not to zero and not only to the total response energy $\sum_i s_i^2$. The practical question is whether throwing away the tail changes prediction performance by an amount that is negligible relative to the uncertainty already left by exact KRR.

Definition 1 (Prediction-risk rank). *For fixed geometry G and tolerance $\alpha > 0$, define*

$$r_\alpha(G) := \min \left\{ r : \sum_{i>r} s_i^2 \leq \alpha R_{\text{KRR}}(G) \right\}.$$

In the experiments below we write $r_T := r_{5\%}$, i.e. $\alpha = 0.05$. This means that r_T is the smallest rank for which an optimally chosen reduced KRR response would have prediction risk at most

$$(1.05) R_{\text{KRR}}(G).$$

Thus r_T is not the algebraic rank of T , and it is not the number of nonzero singular values. It is the number of task-relevant KRR directions needed to be prediction-equivalent to exact KRR at the declared tolerance. Directions beyond r_T may be mathematically nonzero, but their combined effect on target prediction is below the chosen risk budget.

This is a finite-episode benchmark. It is computed from the sampled KRR operator before looking at activations. If $\dim \text{span}(Q) < r_T$, then no rank- $\dim \text{span}(Q)$ reduced KRR operator can reach the five-percent prediction-risk tolerance. If $\dim \text{span}(Q) \geq r_T$, success is still not automatic: the span must align with the leading right singular directions of C_G , because those are the task-label modes that exact KRR uses for prediction.

The same definition applies to nonlinear kernels after replacing K and K_t by the corresponding finite kernel matrices. No population spectral assumption is needed for the experiments below.

Proposition 2 (Prediction-risk excess for T_Q). *Let $u = A^{1/2}z$, $z \sim \mathcal{N}(0, I)$, and let $Q^\top A Q = I$. With*

$$C = K_t A^{-1/2}, \quad W = A^{1/2} Q, \quad P_Q = W W^\top,$$

the reduced-operator excess risk satisfies

$$\rho_G(T_Q) = \frac{\|C(I - P_Q)\|_F^2}{R_{\text{KRR}}(G) + \eta_{\text{num}}}.$$

If W spans the top r right singular directions of C , then

$$\rho_G(T_Q) = \frac{\sum_{i>r} s_i^2}{R_{\text{KRR}}(G) + \eta_{\text{num}}}.$$

Proof. Since $W = A^{1/2} Q$, $W^\top W = Q^\top A Q = I$. For $u = A^{1/2} z$,

$$T u = K_t A^{-1} A^{1/2} z = C z, \quad T_Q u = K_t Q Q^\top A^{1/2} z = K_t A^{-1/2} W W^\top z = C P_Q z.$$

Taking expectation over z gives

$$\mathbb{E} \|(T - T_Q)u\|_2^2 = \|C(I - P_Q)\|_F^2.$$

The singular-value expression follows from Eckart–Young: among all rank- r right projections, the top right singular subspace of C minimizes the Frobenius residual, leaving $\sum_{i>r} s_i^2$. $\square$

Remark 1 (Reading the numbers). *The number $\rho_G(S)$ is an excess-risk ratio. Thus $\rho_G(S) = 0.05$ means that using S instead of exact KRR adds five percent of the KRR posterior prediction risk. Values near zero mean the reduced operator is prediction-equivalent to KRR at the chosen tolerance. Values near or above one mean that the reduction adds error comparable to, or larger than, the irreducible KRR risk. Transformer response errors $E_\varepsilon(F, T)$ and $E_\varepsilon(F, T_Q)$ are different finite-difference diagnostics and are not bounded by this risk scale.*

The mechanism argument therefore needs three distinct checks:

$$E_\varepsilon(F, T; U) \text{ small}, \quad \rho_G(T_Q) \text{ small}, \quad E_\varepsilon(F, T_Q; U) \text{ small}.$$

The first is behavioral. The second says the residual-stream subspace supports ridge in the prediction-risk sense. The third says the transformer’s local response follows the reduced operator induced by that same subspace. The decomposition

$$D_\varepsilon F(y)[u] - Tu = (D_\varepsilon F(y)[u] - T_Q u) + (T_Q u - Tu)$$

keeps these claims separate.

It remains to describe how the residual-stream subspace is extracted. The experiments use two final-state constructions in this section, and the next section introduces the causal layerwise construction. Given any context-pattern matrix $M \in \mathbb{R}^{n_{\text{ctx}} \times p}$, compute the A -weighted SVD

$$A^{1/2}M = U\Sigma V^\top,$$

keep the selected left singular directions, and map back:

$$Q(M) := A^{-1/2}U_{\text{kept}}.$$

Then $Q(M)^\top A Q(M) = I$. The kept singular directions are those above the numerical threshold or within the predeclared rank budget. The prediction-risk rank tells how many ridge-relevant directions should be enough in principle; the experiments then ask whether the residual-stream directions achieve small $\rho_G(T_Q)$ around that scale.

The raw final-state extractor uses the unperturbed final residual stream,

$$M_{\text{raw}} = H_L^{\text{ctx}}, \quad Q_{\text{raw}} := Q(M_{\text{raw}}).$$

It asks an availability question: which context-position directions are linearly visible in the final residual stream, and are enough of them present to support the prediction-relevant ridge operator?

The response-only extractor uses finite-difference changes in the final residual stream,

$$Z_L(u) = \frac{H_L^{\text{ctx}}(y + \varepsilon u) - H_L^{\text{ctx}}(y - \varepsilon u)}{2\varepsilon}, \quad M_{\text{resp}} = [Z_L(u_1), \dots, Z_L(u_m)],$$

and sets

$$Q_{\text{resp}} := Q(M_{\text{resp}}).$$

It asks the stricter question: among the directions that actually move when labels are perturbed, is there a context-position subspace whose induced T_Q has small prediction-risk excess?

In both constructions Q is frozen after extraction. If the basis were re-extracted for each held-out perturbation, it would not define one stable reduced operator. The certificate is that a fixed subspace extracted from residual-stream activations induces a fixed T_Q that agrees with ridge and with the transformer’s held-out local response.

5 Layerwise Availability Construction

The final-state certificate can show that a useful subspace is present at the end of the computation, but it does not explain how that subspace appears across layers. The revised layerwise construction

is a post-training diagnostic for this question. It asks, at each layer, whether the layer’s label-response span contains a subspace of approximately r_T dimensions whose induced reduced KRR operator reaches the prediction-risk tolerance.

This replaces the older primary emphasis on one-at-a-time causal direction selection. Individual ablation is a useful causal-use diagnostic, but it is not the cleanest definition of emergence: a distributed representation can contain the right subspace while no single arbitrary basis direction is individually necessary. The layerwise sufficiency construction first finds a KRR-sufficient subspace; causal intervention should then test the selected subspace as a whole.

At layer ℓ , define the hidden-state finite-difference response

$$Z_\ell(u) := \frac{H_\ell^{\text{ctx}}(y + \varepsilon u) - H_\ell^{\text{ctx}}(y - \varepsilon u)}{2\varepsilon}.$$

Stacking these matrices over build probes gives

$$M_\ell = [Z_\ell(u_1), \dots, Z_\ell(u_m)].$$

The columns of M_ℓ are context-position patterns that are visible at layer ℓ when context labels move. An A -weighted SVD proposes an A -orthonormal candidate response span

$$A^{1/2}M_\ell = U_\ell \Sigma_\ell V_\ell^\top, \quad q_i = A^{-1/2}U_{\ell,i}.$$

Keeping the singular directions above a numerical threshold gives a candidate span C_ℓ . We then order directions inside C_ℓ by KRR usefulness. If C_ℓ is represented by an A -orthonormal matrix, set

$$W_\ell = A^{1/2}C_\ell, \quad Z = K_t A^{-1/2}.$$

The eigenvectors of

$$W_\ell^\top Z^\top Z W_\ell$$

ordered from largest to smallest define nested prefixes

$$Q_{\ell,0} \subset Q_{\ell,1} \subset \dots \subset C_\ell.$$

For each prefix we form

$$T_{Q_{\ell,k}} = K_t Q_{\ell,k} Q_{\ell,k}^\top$$

and evaluate

$$\rho_G(T_{Q_{\ell,k}}) = \frac{\|(T - T_{Q_{\ell,k}})A^{1/2}\|_F^2}{R_{\text{KRR}} + \eta_{\text{num}}}.$$

The two ranks in the experiment have different meanings. The prediction-risk rank $r_T(G; 0.05)$ is intrinsic to the finite KRR task: it is computed from $C_G = K_t A^{-1/2}$ before looking at the transformer. It is the smallest optimal rank that leaves at most five percent of KRR posterior risk in the discarded tail. By contrast, the layerwise rank-to-five-percent

$$k_\ell^{5\%} := \min\{k : \rho_G(T_{Q_{\ell,k}}) \leq 0.05\}$$

is representation-dependent: it asks how many directions from the transformer’s candidate response span C_ℓ are needed to reach the same risk tolerance. If $k_\ell^{5\%} \approx r_T$, then the layer contains an almost optimally aligned KRR subspace. If $k_\ell^{5\%} \gg r_T$, the span contains relevant information but in a poorly aligned form. If no such k exists, that layer’s candidate span is not sufficient. We also report $\rho_G(T_{Q_{\ell,r_T}})$, the excess risk of using exactly the intrinsic task rank, and the best error achievable inside C_ℓ .

6 Experiment 1: Final-State Operator Certificate

The experiments are organized as an evidence ladder. Experiment 1 first certifies that the trained model has a task-local ridge-like response at the final output and that this response is explained by a compact final-state context subspace. Experiment 2 asks where such a subspace becomes available across layers. Experiment 3 then intervenes on the certified directions to test causal use.

The first step is deliberately narrow. For each episode, the geometry G is fixed, exact KRR is the operator $T = K_t A^{-1}$, and the transformer is probed only by infinitesimal context-label perturbations. Build probes are used to extract the raw and response final-state bases; all reported errors use held-out task-label probes after the basis is frozen. The run uses 4 episodes, 16 build probes, 32 held-out probes, $\varepsilon = 10^{-3}$, and extraction rank cap $r_{\max} = 9$.

The endpoint certificate has three parts. First, the model’s local response is already close to KRR on task-label directions. Second, the activation-derived subspace induces a KRR-structured operator $T_Q = K_t Q Q^\top$ whose prediction-risk excess is far below the five-percent tolerance. Third, this is not merely a decodability result: a same-span least-squares readout can fit the model response slightly better, but it is much less KRR-like.

Table 1: Experiment 1 endpoint certificate, mean over 4 episodes. $E(F, S)$ is the held-out finite-difference operator error on task-label probes. $\rho_G(S)$ is exact prediction-risk excess relative to KRR.

Check	Result	Reading
Model vs. exact KRR	$E(F, T) = 0.01207$; pointwise discrepancy 0.01045	The checkpoint is locally KRR-like on task-label perturbations.
Raw final basis	$\rho_G(T_Q) = 0.00279$, $E(F, T_Q) = 0.01195$	The final residual stream contains a highly sufficient KRR source subspace.
Response final basis	$\rho_G(T_Q) = 0.01750$, $E(F, T_Q) = 0.01239$	The sufficient subspace is also visible in label-induced hidden-state motion.
Same-span readout control	raw: $E(F, T_{\text{actLR}}) = 0.00729$, $\rho_G(T_{\text{actLR}}) = 0.16040$	A free readout can fit the model, but the KRR-constrained readout is the one that remains KRR-like.
Matched random subspaces	$\rho_G(T_Q) \approx 15.9$ at raw rank, ≈ 11.7 at response rank	Sufficiency is not explained by rank alone.
Isotropic label probes	$E(F, T) \approx 0.302$	The certificate is task-local, not a global equality with ridge regression.

The finite-difference comparison is meaningful because the local response is close to additive: at $\varepsilon = 10^{-3}$, the normalized additivity defect is 3.96×10^{-4} on task probes and 3.91×10^{-3} on isotropic probes. Thus Experiment 1 establishes the basic endpoint claim: on the task distribution, a compact activation subspace supports the KRR-constrained operator that the transformer locally follows.

7 Experiment 2: Prediction-Risk Rank and Layerwise Availability

Experiment 2 separates two facts that are easy to conflate. The prediction-risk rank r_T is an intrinsic property of the finite KRR problem; it is computed from $C_G = K_t A^{-1/2}$ before looking at the transformer. The layerwise rank $k_\ell^{5\%}$ is representation-dependent; it asks how many directions

from the transformer’s layer- ℓ response span are needed to reach $\rho_G \leq 0.05$. The diagnostic therefore asks not whether the hidden state has dimension r_T , but whether it contains an approximately r_T -dimensional KRR-sufficient subspace.

At each layer, we form the response span C_ℓ from $M_\ell = D_y H_\ell^{\text{ctx}}$. Within that candidate span, directions are ordered by KRR usefulness, and prefixes $Q_{\ell,k}$ are evaluated by

$$\rho_G(T_{Q_{\ell,k}}) = \frac{\|(T - T_{Q_{\ell,k}})A^{1/2}\|_F^2}{R_{\text{KRR}} + \eta_{\text{num}}}.$$

The small linear checkpoints calibrate the diagnostic in the regime where the answer is expected to be simple. For $d_x = 3, 5, 8, 10, 15$, with $n_{\text{ctx}} = 47, n_{\text{tgt}} = 16$, the prediction-risk rank matches the input dimension up to the target cap, and the final rank-to-five-percent matches r_T . This is not the main emergence result: in these linear tasks, the sufficient source subspace is already visible very early. Its role is to show that the revised r_T , $k_\ell^{5\%}$, and ρ_G diagnostic recovers the obvious low-dimensional answer.

Table 2: Experiment 2 linear calibration. The table checks that the prediction-risk diagnostic recovers $r_T \approx d_x$ in simple linear tasks. “First layer” is the first layer whose response span reaches $\rho_G \leq 0.05$.

Checkpoint	r_T	first layer	final cand. dim.	$k_L^{5\%}$	$E(F, T)$
$d_x = 3$	3.0	0	27.5	3.0	0.00617
$d_x = 5$	5.0	0	27.6	5.0	0.00918
$d_x = 8$	8.0	0	40.1	8.0	0.01250
$d_x = 10$	10.0	0	40.4	10.0	0.01433
$d_x = 15$	14.9	2	46.6	14.9	0.01735

The cleanest emergence result is instead the nonlinear RBF checkpoint trained and diagnosed at $n_{\text{ctx}} = 128, n_{\text{tgt}} = 128$. The model itself is close to exact KRR in this regime:

$$E(F, T) = 0.0306, \quad \rho(F) = 0.0203.$$

For this finite KRR problem, $r_T \approx 24.75$. Unlike the small linear sweep, the relevant subspace is not already present at layer 0. It appears over the first few layers.

Table 3: Experiment 2 main emergence result: selected layers of the RBF 128×128 checkpoint. Here $r_T \approx 24.75$. The response span is ordered by KRR usefulness before evaluating ρ_G .

Layer	cand. dim.	rank to 5%	$\rho_G(T_{Q_{\ell,r_T}})$	best excess
0	16	—	1.147	1.147
1	51	38.0	0.100	0.074
2	84	26.9	0.058	0.016
4	107	25.2	0.050	0.004
8	119	24.9	0.048	0.001

The transition is the main point. Layer 0 has only a 16-dimensional response span and is far from sufficient. By layer 2, a nearly task-rank subspace is visible; by layers 4 and 8, $k_\ell^{5\%}$ matches r_T to within finite-sample noise. The final response span is much larger than r_T , but it contains an approximately 25-dimensional KRR-sufficient subspace. Experiment 2 is therefore an availability diagnostic: it locates when the prediction-risk-rank subspace becomes linearly available, without yet claiming that the model causally uses it.

7.1 Stress Cases: Availability Is Not the Same as Use

The $d_x = 40$ checkpoint gives the cleanest warning against over-reading availability. It was trained in a target-poor regime:

$$n_{\text{tgt}} = 8, \quad \tau(\Sigma) = \sum_{i=1}^{40} \frac{\lambda_i}{\lambda_i + \sigma^2} \leq 16,$$

with n_{ctx} sampled as a multiple of the effective dimension. The diagnostic instead uses

$$n_{\text{ctx}} = 47, \quad n_{\text{tgt}} = 40,$$

so it asks for a richer simultaneous target operator than the checkpoint saw directly in one training episode. The checkpoint README reports

$$\text{MSE}/\text{MSE}_{\text{KRR}} = 1.638,$$

so this checkpoint is not an exact-KRR solver. The diagnostic asks whether the error comes from missing source directions or from imperfect use of directions that are present.

Table 4: $d_x = 40$ regime-mismatch stress test. Here $r_T = 21.0$ and $E(F, T) = 0.0939$ for all rows.

Basis	cand. dim.	reach	$\rho_G(T_{Q_{L,r_T}})$	best ρ_G
response	40.0	0.50	0.118	0.086
raw	39.3	1.00	0.047	0.007
combined	40.0	0.75	0.077	0.043

The raw final residual stream contains a KRR-sufficient source subspace at the prediction-risk rank: $\rho_G(T_{Q_{L,r_T}}) \approx 0.047$, just below the five-percent tolerance. The response-only span is close but not reliably sufficient at the same rank, with $\rho_G \approx 0.118$ and reach 0.50. Thus the target-rich failure is not simply absence of the relevant directions. A KRR-oriented subspace is visible in the residual stream, but the label-induced response geometry and the realized model map do not stably use it as the exact target-rich KRR operator.

We also reran the revised layerwise diagnostic on the older fixed- $\ell = 3$ RBF checkpoint `rbf_fixed_13_original.pt`, at $n_{\text{ctx}} = 47$ and $n_{\text{tgt}} = 16, 64, 128$. As n_{tgt} grows, r_T increases $10.6 \rightarrow 17.1 \rightarrow 18.3$, and the final rank-to-five-percent tracks it $11.1 \rightarrow 17.8 \rightarrow 19.3$. But the realized model response degrades $E(F, T) = 0.067 \rightarrow 0.074 \rightarrow 0.097$. This older checkpoint therefore shows the same softer failure mode as $d_x = 40$: a near-sufficient subspace can be present while the model’s realized readout/routing is not as clean as in the matched 128×128 RBF regime.

8 Experiment 3: Causal Direction Removal

The last step tests causal use. Experiments 1 and 2 show that a KRR-sufficient subspace is present; that still does not prove that the transformer relies on it. Experiment 3 therefore takes a certified raw final basis $Q = [q_1, \dots, q_r]$, ranks its directions by KRR task leverage, and removes selected directions from the context-token residual stream before rolling the remaining layers forward.

The leverage score ranks directions by their effect on the KRR-constrained reduced operator, not by activation magnitude. For Q_{-j} , the basis with q_j removed, define

$$\Lambda_j^{\text{red}}(U) := \left(\frac{\sum_{u \in U} \|(T_Q - T_{Q_{-j}})u\|_2^2}{\sum_{u \in U} \|Tu\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

High-leverage directions are the ones whose deletion most changes the reduced KRR response on task-label probes. The intervention removes selected directions at residual state s :

$$H_s^{\text{ctx}} \mapsto H_s^{\text{ctx}} - \sum_{j \in J} q_j q_j^\top A H_s^{\text{ctx}},$$

and compares against matched controls: low-task-leverage directions from the same basis, high-isotropic-but-low-task directions, high-activation-variance-but-low-task directions, and random subsets.

Before surgery, the checkpoint is in the task-local certificate regime:

$$E_{\text{task}}(F, T) = 0.00926, \quad E_{\text{task}}(F, T_Q) = 0.00920, \quad \rho_G(T_Q) = 0.00149.$$

The same episodes retain the off-distribution drift seen in Experiment 1:

$$E_{\text{iso}}(F, T) = 0.266.$$

Table 5: Experiment 3 causal-use summary. Direction removals are at residual state $s = 1$.

Intervention	Result	Reading
No surgery	$E_{\text{task}}(F, T) = 0.00926$, $\rho_G(T_Q) = 0.00149$	The starting episodes satisfy the same task-local certificate.
Remove one high-task-leverage direction	prediction drift 4.68, response damage 41.59, KRR-error inflation $649\times$	Directions selected by KRR leverage are causally important.
Remove one low-task-leverage direction	prediction drift 9.8×10^{-4} , response damage 0.0024, inflation $1.03\times$	Removing same-basis directions with low task leverage is nearly inert.
Keep only the Q -projection	$E_{\text{task}}(F, T) = 0.01836$, response damage 0.01634	The certified subspace nearly preserves the task-local response by itself.
Remove Q	$E_{\text{task}}(F, T) = 106.46$, response damage 106.46	The complement does not carry the task-local KRR response.

The layer sweep adds localization: high-task surgery is most damaging at early residual states, before later attention has routed the relevant context information into target tokens, while context-only surgery after the final layer is a no-op. At $s = 1$, aggregate task leverage and task response damage are strongly related across matched non-random controls (log-log Pearson 0.91, Spearman 0.87).

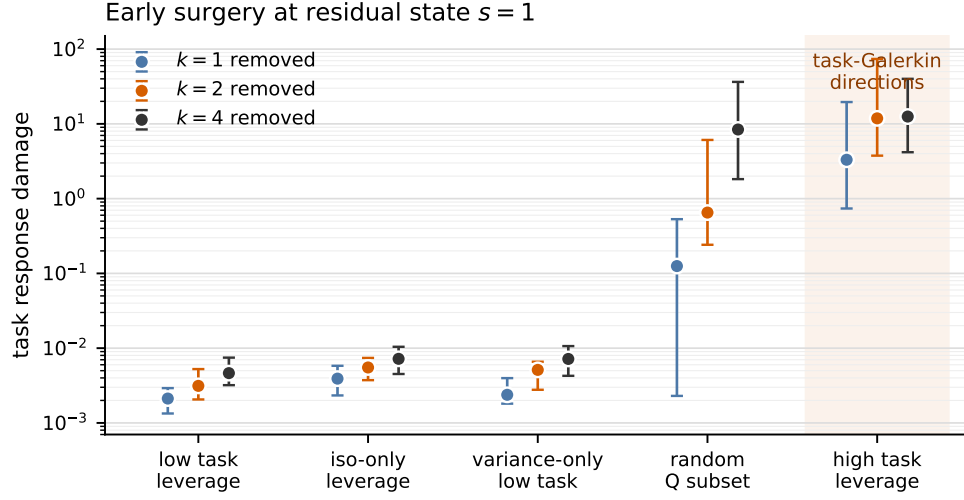


Figure 1: Experiment 3: selective causal effect of task-important activation directions. The plotted metric is task finite-difference response damage. High task-leverage removals cause orders of magnitude larger damage than matched low-task-leverage controls.

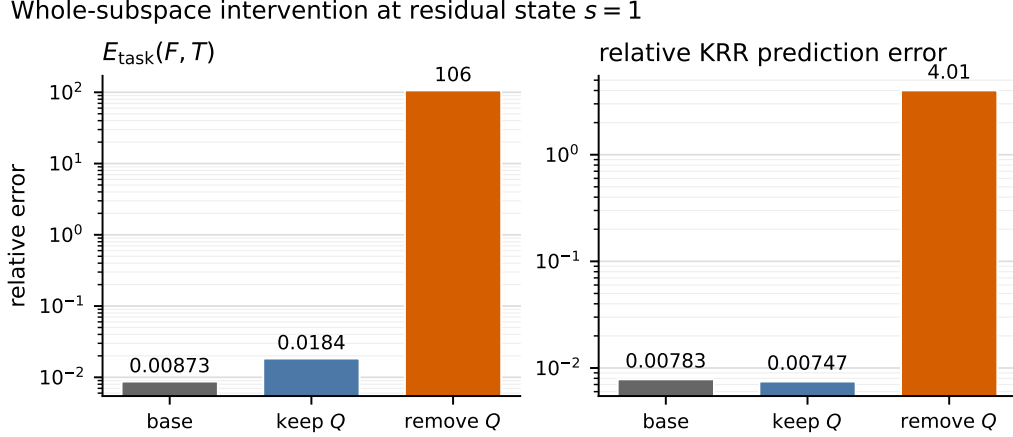


Figure 2: Experiment 3: whole-subspace complement ablation at residual state $s = 1$. Keeping only the Q -projection nearly preserves the task-local response; removing Q destroys it.

Together, the selective removals and complement ablation support the causal-use claim in an operational sense. The directions that the reduced KRR operator identifies as task-important are not merely decodable; preserving them largely preserves the task-local response, and removing them selectively destroys it.

9 What Is Justified

The experiments justify a bounded mechanistic claim:

For the tested checkpoint, geometries, perturbation scale, and task-label probe distribution, the transformer’s local label response is well approximated by a reduced KRR operator induced by a compact context-side activation subspace. In successful settings, layerwise response spans contain a prediction-risk-rank subspace whose induced reduced operator is close to exact KRR; in the original linear setting, high-task-leverage directions from this subspace are causally important for the transformer’s response.

They do not justify stronger claims. They do not show that the transformer globally equals ridge regression. They do not show that derivative matching at one sampled label vector implies equality on a neighborhood. They do not show that the transformer literally executes the explicit algorithm

$$y \mapsto QQ^\top y \mapsto K_t QQ^\top y$$

as its internal program. They also do not show that architecture alone guarantees the behavior. The prediction-risk rank calculation supplies a requirement for context-side subspaces; it is not an SGD theorem.

The clean final formulation is:

The trained transformer locally realizes the ridge-regression label-response operator on task-distributed directions. In the tested regime, this behavior is explained by a compact context-side activation subspace whose induced reduced ridge operator approximates exact KRR at the prediction-risk rank. The layerwise diagnostic tracks when such a subspace becomes available; the causal intervention then asks whether the model actually uses the selected directions.

10 Logical Chain

The argument can be read as a sequence of commitments, each narrower than the previous one:

1. Freeze the episode geometry $G = (X_{\text{ctx}}, X_{\text{tgt}})$.
2. Under fixed G , ridge regression becomes $T = K_t A^{-1}$.
3. Treat the transformer as a label map $F : \mathbb{R}^{n_{\text{ctx}}} \rightarrow \mathbb{R}^{n_{\text{tgt}}}$.
4. Estimate the local response $D_\epsilon F(y)[u]$.
5. Test behavior by checking $E_\epsilon(F, T)$ on task-label probes.
6. Compute the prediction-risk rank r_T from $C_G = K_t A^{-1/2}$ and R_{KRR} .
7. Extract activation-derived context-side candidate spans C_ℓ .

8. Order each candidate span by KRR-targeted energy and form prefixes $Q_{\ell,k}$.
9. Construct the KRR-constrained reduced operator $T_Q = K_t Q Q^\top$.
10. Test subspace sufficiency with $\rho_G(T_Q)$.
11. Test model-subspace alignment with $E_\varepsilon(F, T_Q)$.
12. In causal runs, remove high-leverage Q -directions and check whether task-local response is damaged.

This is the full mechanism story: behavioral similarity, a KRR-structured activation certificate, prediction-risk-rank sufficiency, layerwise emergence of the relevant subspace, and causal specificity of high-leverage directions where the intervention has been validated.